ACADEMIC PROJECT TIMELINE

Lesion-Aware Open-Set Plant Disease Recognition

Extension of ACM MM'24 MVPDR | December 2025-January 2026

December 2025 - foundation and lesion-aware recognition

DEC 2025 | EARLY

Scope and evaluation design

- Defined the student-scale MVPDR extension and its four-component boundary.
- Set the few-shot lab-to-field, held-out-class, and validation-only evaluation contract.

DEC 2025 | MID

CLIP baseline and frozen DINOv3 local features

- Structured textual and global visual prototype evidence.
- Added dense local features for lesion texture, colour, and shape with cache provenance.

DEC 2025 | LATE

PlantSeg-supervised lesion decoder

- Trained a lightweight decoder with BCE + Dice supervision over frozen patch features.
- Pooled classification features using predicted masks; test masks remained evaluation-only.

January 2026 - fusion, open-set evaluation, and consolidation

JAN 2026 | EARLY

Three-view gated fusion

- Fused textual, global, and lesion-localized prototype evidence with a per-image gate.
- Preserved a parameter-efficient design: 0.8% trainable parameters.

JAN 2026 | MID

Calibration and held-out disease detection

- Added temperature scaling and validation-selected open-set rejection.
- Evaluated 20-shot lab-to-field recognition and held-out disease detection.

JAN 2026 | LATE

Analysis and academic project handoff

- Consolidated the A-E experimental progression and documented method boundaries.
- Closed the academic project period in January 2026.

PROJECT SUMMARY

From baseline to open-set recognition

The project advanced through five controlled stages, with each stage adding one testable capability while keeping both foundation encoders frozen.

A	MVPDR-style baseline CLIP text + global prototypes
B	Local representation + frozen DINOv3 patch features
C	Lesion supervision + predicted-lesion feature pooling
D	Three-view fusion + learned per-image evidence gate
E	Open-set recognition + calibrated unknown rejection

Historical outcomes

0.8%	**0.76**	**67%**	**+6 pp**	**0.90**	**40%**
trainable parameters	mean binary-lesion Dice	20-shot lab-to-field macro-F1	vs same-split MVPDR reproduction	open-set AUROC	FPR95

Evaluation safeguards

Leakage control Training samples build prototypes; validation selects temperature and rejection thresholds; target-domain test data is evaluated once.	Mask integrity Ground-truth masks supervise and evaluate the decoder. Classification inference pools only under predicted lesion masks.
Held-out diseases Unknown classes are excluded from training prototypes and represented in validation and test for open-set calibration and evaluation.	Student-scale scope CLIP and DINOv3 remain frozen; only the lightweight decoder, evidence gate, and calibration components are trained.

> Provenance. These are author-supplied historical results from the December 2025-January 2026 academic project. The current Git repository is a later reproducibility reconstruction and has not independently rerun the corresponding real-data experiments. This phase timeline is not a fabricated commit or activity log.